

- domain of streptococcal protein G and comparison with NMR. *Biochemistry*, 33(15):4721–4729, 1994.
- Gao, L., Dupré la Tour, T., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., and Wu, J. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025.
- Garcia, E. N. V. and Ansuini, A. Interpreting and steering protein language models through sparse autoencoders. In *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design*, 2025.
- Gronenborn, A. M., Filpula, D. R., Essig, N. Z., Achari, A., Whitlow, M., Wingfield, P. T., and Clore, G. M. A novel, highly stable fold of the immunoglobulin binding domain of streptococcal protein G. *Science*, 253(5020):657–661, 1991.
- Gujral, O., Bafna, M., Alm, E., and Berger, B. Sparse autoencoders uncover biologically interpretable features in protein language model representations. *Proceedings of the National Academy of Sciences*, 122(34):e2506316122, 2025.
- Hanukoglu, I. Proteopedia: Rossmann fold: A beta-alpha-beta fold at dinucleotide binding sites. *Biochemistry and Molecular Biology Education*, 43(3):206–209, 2015.
- Hayes, T., Rao, R., Akin, H., Sofroniew, N. J., Oktay, D., Lin, Z., Verkuil, R., Tran, V. Q., Deaton, J., Wiggert, M., Badkundri, R., Shafkat, I., Gong, J., Derry, A., Molina, R. S., Thomas, N., Khan, Y. A., Mishra, C., Kim, C., Bartie, L. J., Nemeth, M., Hsu, P. D., Sercu, T., Candido, S., and Rives, A. Simulating 500 million years of evolution with a language model. *Science*, 387(6736):850–858, 2025.
- Huang, L.-K., Zhu, R., He, B., and Yao, J. Steering protein language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267, pp. 26247–26260. PMLR, 2025.
- Kornev, A. P., Haste, N. M., Taylor, S. S., and Ten Eyck, L. F. Surface comparison of active and inactive protein kinases identifies a conserved activation mechanism. *Proceedings of the National Academy of Sciences*, 103(47):17783–17788, 2006.
- Krzus, M. Replicating the circuit kings. https://www.mattkrus.com/posts/circuit_tracing/, 2025.
- Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., Shmueli, Y., dos Santos Costa, A., Fazel-Zarandi, M., Sercu, T., Candido, S., and Rives, A. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- Makhzani, A. and Frey, B. k -sparse autoencoders. *arXiv preprint arXiv:1312.5663*, 2013.
- Merullo, J., Watts, C., Loeffler, M., Gorton, L., Simon, E., McGrath, T., and Lewis, O. Replicating circuit tracing for a simple known mechanism. Goodfire Research, 2025. URL <https://www.goodfire.ai/research/replicating-circuit-tracing-for-a-simple-mechanism#>.
- Modi, V. and Dunbrack, R. L. Defining a new nomenclature for the structures of active and inactive kinases. *Proceedings of the National Academy of Sciences*, 116(14):6818–6827, 2019.
- Nainani, J., Reimer, B. M., Watts, C., Jensen, D., and Green, A. G. Mechanistic evidence that motif-gated domain recognition drives contact prediction in protein language models. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.
- Notin, P., Kollasch, A., Ritter, D., van Niekerk, L., Paul, S., Spinner, H., Rollins, N., Shaw, A., Orenbuch, R., Weitzman, R., Frazer, J., Dias, M., Franceschi, D., Gal, Y., and Marks, D. ProteinGym: Large-scale benchmarks for protein fitness prediction and design. In *Advances in Neural Information Processing Systems*, volume 36, pp. 64331–64379, 2023a.
- Notin, P., Weitzman, R., Marks, D., and Gal, Y. Protein-NPT: Improving protein property prediction and design with non-parametric transformers. In *Advances in Neural Information Processing Systems*, volume 36, pp. 33529–33563, 2023b.
- Ohgami, R. S., Campagna, D. R., Greer, E. L., Antiochos, B., McDonald, A., Chen, J., Sharp, J. J., Fujiwara, Y., Barker, J. E., and Fleming, M. D. Identification of a ferriredutase required for efficient transferrin-dependent iron uptake in erythroid cells. *Nature Genetics*, 37(11):1264–1269, 2005.
- Olson, C. A., Wu, N. C., and Sun, R. A comprehensive biophysical description of pairwise epistasis throughout an entire protein domain. *Current Biology*, 24(22):2643–2651, 2014.
- Parsan, N., Yang, D. J., and Yang, J. J. Towards interpretable protein structure prediction with sparse autoencoders. In *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design*, 2025.
- Paysan-Lafosse, T., Blum, M., Chuguransky, S., Grego, T., Pinto, B. L., Salazar, G. A., Bileschi, M. L., Bork, P., Bridge, A., Colwell, L., Gough, J., Haft, D. H., Letunić, I., Marchler-Bauer, A., Mi, H., Natale, D. A., Orengo, C. A., Pandurangan, A. P., Rivoire, C., Sigrist, C. J. A.,

- Sillitoe, I., Thanki, N., Thomas, P. D., Tosatto, S. C. E., Wu, C. H., and Bateman, A. Interpro in 2022. *Nucleic Acids Research*, 51(D1):D418–D427, 2022.
- Pokusaeva, V. O., Usmanova, D. R., Putintseva, E. V., Espinar, L., Sarkisyan, K. S., Mishin, A. S., Bogatyreva, N. S., Ivankov, D. N., Akopyan, A. V., Avvakumov, S. Y., Povolotskaya, I. S., Fillion, G. J., Carey, L. B., and Kondrashov, F. A. An experimental assay of the interactions of amino acids from orthologous sequences shaping a complex fitness landscape. *PLOS Genetics*, 15(4): e1008079, 2019.
- Rimsky, N., Gabrieli, N., Schulz, J., Tong, M., Hubinger, E., and Turner, A. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2024.
- Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., Guo, D., Ott, M., Zitnick, C. L., Ma, J., and Fergus, R. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15), 2021.
- Sarkisyan, K. S., Bolotin, D. A., Meer, M. V., Usmanova, D. R., Mishin, A. S., Sharonov, G. V., Ivankov, D. N., Bozhanova, N. G., Baranov, M. S., Soylemez, O., Bogatyreva, N. S., Vlasov, P. K., Egorov, E. S., Logacheva, M. D., Kondrashov, A. S., Chudakov, D. M., Putintseva, E. V., Mamedov, I. Z., Tawfik, D. S., Lukyanov, K. A., and Kondrashov, F. A. Local fitness landscape of the green fluorescent protein. *Nature*, 533(7603):397–401, 2016.
- Sauer-Eriksson, A., Kleywegt, G. J., Uhlén, M., and Jones, T. Crystal structure of the C2 fragment of streptococcal protein G in complex with the Fc domain of human IgG. *Structure*, 3(3):265–278, 1995.
- Seuma, M., Lehner, B., and Bolognesi, B. An atlas of amyloid aggregation: the impact of substitutions, insertions, deletions and truncations on amyloid beta fibril nucleation. *Nature Communications*, 13(1), 2022.
- Simon, E. and Zou, J. InterPLM: Discovering interpretable features in protein language models via sparse autoencoders. *Nature Methods*, 22(10):2107–2117, 2025.
- Sinai, S., Jain, N., Church, G. M., and Kelsic, E. D. Generative AAV capsid diversification by latent interpolation. *bioRxiv*, 2021.
- Sloan, D. J. and Hellinga, H. W. Dissection of the protein G B1 domain binding site for human IgG Fc fragment. *Protein Science*, 8(8):1643–1648, 1999.
- Steinegger, M. and Söding, J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11):1026–1028, 2017.
- Suzek, B. E., Huang, H., McGarvey, P., Mazumder, R., and Wu, C. H. UniRef: Comprehensive and non-redundant UniProt reference clusters. *Bioinformatics*, 23(10):1282–1288, 2007.
- Taylor, S. S. and Kornev, A. P. Protein kinases: Evolution of dynamic regulatory proteins. *Trends in Biochemical Sciences*, 36(2):65–77, 2011.
- Templeton, A., Conerly, T., Marcus, J., Lindsey, J., Bricken, T., Chen, B., Pearce, A., Citro, C., Ameisen, E., Jones, A., Cunningham, H., Turner, N. L., McDougall, C., MacDiarmid, M., Freeman, C. D., Summers, T. R., Rees, E., Batson, J., Jermyn, A., Carter, S., Olah, C., and Henighan, T. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Transformer Circuits Thread, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- Tsuboyama, K., Dauparas, J., Chen, J., Laine, E., Mohseni Behbahani, Y., Weinstein, J. J., Mangan, N. M., Ovchinnikov, S., and Rocklin, G. J. Mega-scale experimental analysis of protein folding stability in biology and design. *Nature*, 620(7973), 2023.
- Tsui, D. and Aghazadeh, A. On recovering higher-order interactions from protein language models. In *ICLR 2024 Workshop on Generative and Experimental Perspectives for Biomolecular Design*, 2024.
- Tsui, D., Musharraf, A., Erginbas, Y. E., Kang, J. S., and Aghazadeh, A. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Advances in Neural Information Processing Systems*, 2025a.
- Tsui, D., Talreja, K., and Aghazadeh, A. Sparse autoencoders for low- N protein function prediction and design. In *NeurIPS 2025 AI for Science Workshop*, 2025b.
- Walton, T., Tsui, D., Fogel, L., Huard, D. J. E., Chagas, R. S., Lieberman, R. L., and Aghazadeh, A. GOLF: A generative AI framework for pathogenicity prediction of myocilin OLF variants. In *Proceedings of the 20th Machine Learning in Computational Biology Meeting*, volume 311 of *Proceedings of Machine Learning Research*, pp. 148–161. PMLR, 2025a.
- Walton, T., Tsui, D., Musharraf, A., and Aghazadeh, A. SpecMER: Fast protein generation with K-mer guided speculative decoding. In *Advances in Neural Information Processing Systems*, 2025b.

Weng, C., Faure, A. J., Escobedo, A., and Lehner, B. The energetic and allosteric landscape for KRAS inhibition. *Nature*, 626(7999):643–652, 2023.

Wu, Z., Kan, S. B. J., Lewis, R. D., Wittmann, B. J., and Arnold, F. H. Machine learning-assisted directed protein evolution with combinatorial libraries. *Proceedings of the National Academy of Sciences*, 116(18):8852–8858, 2019.